



PLAYFAIR CIPHER TECHNIQUE

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INTRODUCTION

- The Playfair Cipher is a **manual symmetric encryption technique**.
- It was invented by **Charles Wheatstone** in **1854**.
- It was promoted by **Lord Playfair**, after whom it is named.
- It encrypts **pairs of letters (digraphs)** instead of single letters.
- It provides better security than the Caesar Cipher and Monoalphabetic Cipher.



Sir Charles Wheatstone



Sir Lord Playfair

- **Aim:**

To securely encrypt and decrypt text using pairs of letters

- **Algorithm**

1. Choose a keyword.
2. Create a **5 × 5 matrix** using the keyword.
3. Combine **I and J** into one cell.
4. Fill the remaining letters of the alphabet
5. Divide the plaintext into pairs.
6. Insert **X** if two letters in a pair are the same.

Encrypt using these rules:

- Same row → Take the next letter to the right.
- Same column → Take the letter below.
- Rectangle → Replace each letter with the letter in the same row but the other letter's column.

EXAMPLE

Plaintext : HELLO

Split into pairs : HE LX LO

Encryption

- HE $\rightarrow$ CF
- LX $\rightarrow$ SU
- LO $\rightarrow$ PM
- CFSUPM

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

ADVANTAGES

- More secure than simple substitution ciphers.
- Encrypts two letters at a time.
- Easy to implement manually.
- Reduces frequency analysis attacks.
- No complex calculations.

DISADVANTAGES

- Limited to a 5×5 matrix. I and J are combined.
- Vulnerable to modern cryptanalysis.
- Slower for long messages.
- Not suitable for today's secure communication.

APPLICATIONS

- Military communication (historically).
- Educational cryptography .
- Learning encryption techniques .
- Puzzle and coding competitions.